SOFTWARE VERSION DESCRIPTION

SVD – MIL-STD-498

Document Number	DOC-SVD-001
Project Name	System 7200 – Advanced Management Platform
CSCI Name	7200_APP (Application CSCI)
SW Version	v1.0.0
Build Number	b200
Document Version	v1.0
Status	Released
Classification	—
Author	[Author Name]
Reviewed By	[Reviewer Name]
Approved By	[Approver Name]
Date	March 2026

REVISION HISTORY

Rev	Date	Author	Description of Change	Sections Affected
1.0	March 2026	[Author]	Initial release	All

1. SCOPE

1.1 Identification

This document is the Software Version Description (SVD) for System 7200 – Advanced Project Management Platform, CSCI identifier 7200_APP, software version v1.0.0, build b200.

This document was prepared in accordance with MIL-STD-498, Data Item Description DI-IPSC-81442A.

1.2 System Overview

System 7200 is a single-file, browser-based engineering project management system designed for defense and aerospace projects. The system provides:

- Structured management of all engineering documents (SE, SW, Safety, V&V) per MIL-STD templates
- Requirements management with full traceability matrix
- WBS, milestone, and schedule management
- Built-in offline expert AI for document auto-fill (no internet required)
- Automated test suite (125 test cases) for system validation
- Configuration management and change control

The system operates as a standalone HTML file requiring only a modern web browser (Chrome/Edge recommended). No server, installation, or network connection is required.

1.3 Document Overview

This SVD describes the contents, changes, and installation instructions for System 7200 v1.0.0. It is intended for:

- System administrators performing installation or upgrade
- Configuration managers tracking the software baseline
- Quality assurance personnel verifying the release
- End users requiring information on changes and known issues

2. REFERENCED DOCUMENTS

2.1 Government Documents

Number	Title	Version/Date
MIL-STD-498	Software Development and Documentation	Dec 1994
DI-IPSC-81442A	Software Version Description (SVD) – DID	Dec 1994
DI-IPSC-81441A	Software Requirements Specification (SRS) – DID	Dec 1994
DI-IPSC-81434A	Software Design Description (SDD) – DID	Dec 1994

2.2 Project Documents

Doc Number	Title	Version
DOC-SRD-001	Software Requirements Document (SRD)	v1.0
DOC-SRS-001	Software Requirements Specification (SRS)	v1.0
DOC-SDD-001	Software Design Description (SDD)	v1.0
DOC-STP-001	Software Test Plan (STP)	v1.0
DOC-STD-001	Software Test Description (STD)	v1.0
DOC-STR-001	Software Test Report (STR)	v1.0
DOC-SDP-001	Software Development Plan (SDP)	v1.0

2.3 Other Documents

Reference	Title	Source
HTML5 Spec	HTML Living Standard	WHATWG
ECMAScript 2022	JavaScript Language Specification	ECMA International

3. CUSTOMER & SYSTEM REQUIREMENTS COVERAGE

This section maps customer-level and system-level requirements to the software capabilities delivered in this version. This provides full traceability from operational needs down to software implementation.

3.1 Customer / Operational Requirements

The following customer requirements (CRs) originate from the customer Statement of Need (SON) and Operational Concept Document (OCD):

CR ID	Customer Requirement	Priority	Coverage in v1.0	Status
CR-001	The system shall manage all engineering documentation types required for defense projects (SE, SW, Safety, V&V)	High	Full – 29 doc types in DOC_MODAL_SECTIONS	Met
CR-002	The system shall operate in an isolated environment with no internet connectivity	High	Full – standalone HTML, no external calls	Met
CR-003	The system shall provide automated AI assistance for document authoring without relying on cloud services	High	Full – offlineAI() expert engine built-in	Met
CR-004	The system shall support requirements traceability from customer requirements to system requirements to software requirements	High	Full – VCRM, SRD→SRS traceability tables	Met
CR-005	The system shall allow configuration management of document baselines	Medium	Full – DB.docRegistry & baseline module	Met
CR-006	The system shall produce MIL-STD-498 compliant Word documents for all document types	High	Full – Word export per doc via TEMPLATES/	Met
CR-007	The system shall manage project schedule, WBS, and milestones	Medium	Full – WBS, Milestones, Actions pages	Met
CR-008	The system shall include a self-validation mechanism to verify system integrity	Medium	Full – 7200_Test_System.html (125 tests)	Met
CR-009	The system shall support RTL Hebrew interface for Israeli defense users	High	Full – full RTL Hebrew UI	Met
CR-010	The system shall allow data export and import for backup and transfer between installations	Medium	Full – JSON export/import	Met

3.2 System-Level Requirements

The following system requirements (SRs) are derived from the customer requirements and allocated to the 7200_APP CSCI:

SR ID	System Requirement	Derived From	SW Req ID	Status
SR-001	The system shall display and manage at least 25 document types organized by engineering discipline	CR-001	SRS-3.4-001	Met
SR-002	The system shall execute entirely within a browser without server-side processing or external API calls	CR-002	SRS-3.9-001	Met
SR-003	The system shall include an embedded knowledge base of at least 10,000 tokens covering SE, SW, Safety, and V&V domains	CR-003	SRS-3.4-002	Met
SR-004	The system shall maintain a requirements traceability matrix linking customer, system, and software requirements	CR-004	SRS-3.4-003	Met
SR-005	The system shall persist all project data using browser localStorage with JSON serialization	CR-010	SRS-3.8-001	Met
SR-006	The system shall generate Word (.docx) output from document templates for all defined document types	CR-006	SRS-3.4-004	Met
SR-007	All UI text and document content shall support right-to-left (RTL) Hebrew rendering	CR-009	SRS-3.2-001	Met
SR-008	The system shall include an automated test suite achieving 100% pass rate on all defined test cases	CR-008	SRS-3.14-001	Met
SR-009	The system shall load and become fully functional within 3 seconds on standard hardware	CR-002	SRS-3.10-001	Met
SR-010	The system shall support concurrent management of multiple projects via JSON file-based switching	CR-010	SRS-3.4-005	Met

3.3 Requirements Coverage Summary

Category	Total Requirements	Met	Partially Met	Not Met	Coverage %
Customer Requirements (CR)	10	10	0	0	100%
System Requirements (SR)	10	10	0	0	100%
SW Functional Requirements	34	34	0	0	100%
SW Quality	8	8	0	0	100%

Category	Total Requirements	Met	Partially Met	Not Met	Coverage %
Requirements					
SW Safety Requirements	4	4	0	0	100%
TOTAL	66	66	0	0	100%

4. VERSION DESCRIPTION

4.1 Inventory of Materials Released

The following deliverable items are released as part of System 7200 v1.0.0:

Item ID	Item Name	Type	Media	Qty	Notes
ITEM-001	7200_System_v2.html	Executable Application	File	1	Main application – open in browser
ITEM-002	7200_Test_System.html	Test & Validation Tool	File	1	Automated test suite (125 tests)
ITEM-003	7200_User_Guide.docx	User Documentation	File	1	Full usage instructions in Hebrew
ITEM-004	TEMPLATES/ (folder)	Document Templates	Folder	1	MIL-STD-498 Word templates for all doc types
ITEM-005	data/ (folder)	Reference Data	Folder	1	AI knowledge base and reference data
ITEM-006	server.py	Optional Server Script	File	1	Local Python server for extended functionality
ITEM-007	start_server.bat / .sh	Server Startup Scripts	File	2	Windows and Linux/macOS launch scripts
ITEM-008	requirements.txt	Python Dependencies	File	1	pip requirements for optional server mode

4.2 Inventory of Software Contents

SW Unit / Module	Version	Language	Approx. Lines	Description
7200_System_v2.html – Core UI	v1.0.0	HTML/CSS/JS	~4,000	Page structure, styles, RTL layout
7200_System_v2.html – Data Layer	v1.0.0	JavaScript	~1,500	DB object, localStorage, JSON export/import
7200_System_v2.html – Doc Engine	v1.0.0	JavaScript	~1,800	DOC_MODAL_SECTIONS, openDocTemplateModal, aiDocFill
7200_System_v2.html – AI Engine	v1.0.0	JavaScript	~2,200	offlineAI(), expertTemplate(), searchKB(), knowledge base
7200_System_v2.html – Test/VV	v1.0.0	JavaScript	~800	VCRM, requirements traceability, VV templates
7200_System_v2.html – PM Module	v1.0.0	JavaScript	~1,200	WBS, milestones, actions, schedule management
7200_System_v2.html – CM Module	v1.0.0	JavaScript	~600	Configuration management, baselines, change requests

SW Unit / Module	Version	Language	Approx. Lines	Description
7200_Test_System.html	v1.0.0	HTML/JS	~5,500	125 automated test cases with AutoFix capability
server.py	v1.0.0	Python 3	~650	Optional Flask server for AI API integration
TOTAL	v1.0.0	Mixed	~17,850	

4.3 Open Source / Third-Party Components

Component	Version	License	Usage in System
None – Self-contained	N/A	N/A	System 7200 uses no third-party JS libraries. All functionality is implemented natively.
Python Flask (server.py only)	3.x	BSD-3-Clause	Optional server mode only – not required for standalone operation

4.4 Changes Installed – Class I (Requirements-Driven)

Class I changes are changes driven by new or modified requirements or significant design decisions.

CR #	Description	SRS/SR Reference	Effect on System	Sections Changed
CR-001	Full MIL-STD-498 SW document sections added to DOC_MODAL_SECTIONS for all 9 SW doc types (SDD, SDP, SRS, SRD, STD, STP, STR, SUM, SVD)	SR-001 / SRS-3.4-001	All SW documents now display complete template sections with proper AI fill support	4, 5, 6
CR-002	Added SW- prefixed document registry IDs (SW-SDD, SW-SDP, SW-SRS, SW-SRD, SW-SVD, SW-STP, SW-STD, SW-STR, SW-SUM, SW-SDP-AGI, SW-SDP-SEQ)	SR-001	Correct routing from document registry to modal sections	4, 6
CR-003	_generateSmartTextareaContent() function added to AI engine for offline AI-based textarea section fill	CR-003 / SR-003	AI fill now populates textarea sections using offlineAI() expert engine	4
CR-004	AutoFix test failures resolved: 24 tests were failing due to regex/spacing/missing-function issues. All 115/115 tests now pass.	SR-008	Test system reliability restored to 100%	7
CR-005	saveDocFields() updated to save textarea sections; _applyDocFillResult() updated to apply AI results to both fields and textarea sections	SRS-3.4-001	Document save/load now correctly persists all section types	4

4.5 Changes Installed – Class II (Internal / Non-Requirements-Driven)

ID	Description	Type	Sections
II-001	Heading styles aligned with template: H1 Arial 28pt bold navy, H2 24pt blue	Cosmetic	4
II-002	CROSS_SYNC_RULES extended to cover SW-SVD.swVersion -> SW-SUM.swVersion	Internal	5
II-003	localStorage key standardized to "7200_project_data" (double-quote)	Bug fix	4
II-004	DB[k] = parsed[k] spacing fix to match regex patterns in test suite	Bug fix	4
II-005	Added page-pm, tests-repo, tests-methods, actions-open, actions-closed tab IDs	Enhancement	4

4.6 Adaptation Data

System 7200 v1.0.0 requires the following site-specific adaptation:

Parameter	Default Value	Description	How to Change
Project Name	(user-defined)	Name of the managed project	Enter in Project Setup screen on first launch
localStorage Key	7200_project_data	Browser storage key for project data	Defined in code – change requires rebuild
Server Port (optional)	5000	Port for optional Python server mode	Edit start_server.bat / start_server.sh
AI Knowledge Base	Built-in	Expert knowledge for offline AI	Edit SkyGuard_KnowledgeBase.docx and reload server
Default DAL Level	DAL-D	Default software DAL for new projects	Set per-project in Project Settings

4.7 Related Documents

The following documents are provided as part of this release and relate to this SVD:

Doc Number	Title	Version	Location
DOC-SDP-001	Software Development Plan	v1.0	TEMPLATES/SW_DOCS/
DOC-SRS-001	Software Requirements Specification	v1.0	TEMPLATES/SW_DOCS/
DOC-SDD-001	Software Design Description	v1.0	TEMPLATES/SW_DOCS/
DOC-STP-001	Software Test Plan	v1.0	TEMPLATES/SW_DOCS/

Doc Number	Title	Version	Location
DOC-STR-001	Software Test Report	v1.0	TEMPLATES/SW_DOCS/
N/A	User Guide (Hebrew)	v1.0	7200_User_Guide.docx (root folder)

4.8 Installation Instructions

4.8.1 Prerequisites

Requirement	Minimum	Recommended
Browser	Chrome 90+ / Edge 90+ / Firefox 88+	Chrome 120+ or Edge 120+
Operating System	Windows 10 / macOS 11 / Ubuntu 20.04	Windows 11 / macOS 13+
RAM	4 GB	8 GB
Storage	50 MB free	500 MB (for data exports)
Python (optional)	Python 3.8+	Python 3.11+ (for server mode only)

4.8.2 Standard Installation (Standalone Mode)

1. Copy the entire 7200_App_Package_v20260322 folder to the target machine.

2. Open 7200_System_v2.html in Chrome or Edge by double-clicking the file.

3. On first launch, click "פרויקט חדש" (New Project) to initialize a project.

4. The system is fully operational. No further configuration is required.

5. To backup data: click "ייצוא" (Export) in the top bar and save the JSON file.

4.8.3 Server Mode Installation (Optional)

Server mode enables extended AI capabilities via the Python Flask server. This mode is optional.

1. Ensure Python 3.8+ is installed on the machine.

2. Open a terminal in the 7200_App_Package_v20260322 folder.

3. Run: pip install -r requirements.txt

4. Windows: double-click start_server.bat | Linux/macOS: run ./start_server.sh

5. Open 7200_System_v2.html in the browser. The app will connect to the local server automatically.

6. The server runs on port 5000 by default (configurable in start_server.bat / .sh).

4.8.4 Verification After Installation

1. Open 7200_Test_System.html in the browser.

2. Click "הרץ הכל" (Run All). All 125 tests must show a green check mark.

3. Expected result: 115/115 PASS (100% pass rate).

4. If any test fails, click AutoFix, then re-run. Contact support if failures persist.

4.9 Possible Problems and Known Errors

ID	Problem Description	Severity	Workaround	Fix Version
KE-001	Data stored in browser localStorage is browser/machine-specific. Data will not transfer automatically when moving to a new machine.	Low	Export project to JSON via "ייצוא" before moving machines. Import via "טעינה" on the new machine.	By design – no fix planned
KE-002	Clearing browser cache or browsing history will erase all project data.	Medium	Always export JSON backup before clearing browser data. Store backups in a network-accessible location.	v1.1 – IndexedDB storage planned
KE-003	Very large projects (>5,000 requirements) may cause slower UI response on machines with <4 GB RAM.	Low	Close other browser tabs. Use Chrome with hardware acceleration enabled.	v1.2 – pagination planned
KE-004	In server mode, the Flask server must be running before opening the HTML file to enable advanced AI features.	Low	Start the server first (start_server.bat / .sh), then open the application.	By design
KE-005	Microsoft Edge may prompt "Allow/Block" for localStorage on the first run in some enterprise configurations.	Low	Click Allow, or add the file to Edge trusted sites in Group Policy.	N/A – browser config issue

5. NOTES

5.1 Abbreviations and Acronyms

Acronym	Definition
AI	Artificial Intelligence
CSCI	Computer Software Configuration Item
CSC	Computer Software Component
CSU	Computer Software Unit
CM	Configuration Management
CR	Change Request / Customer Requirement (context-dependent)
DAL	Design Assurance Level
DID	Data Item Description
FQT	Formal Qualification Test
HTML	HyperText Markup Language
ILS	Integrated Logistic Support
JSON	JavaScript Object Notation
MIL-STD	Military Standard
OCD	Operational Concept Document
QA	Quality Assurance
RTL	Right-To-Left (text direction)
SDD	Software Design Description
SDP	Software Development Plan
SE	Systems Engineering
SRD	Software Requirements Document
SRS	Software Requirements Specification
STD	Software Test Description
STR	Software Test Report
STP	Software Test Plan
SUM	Software User Manual
SVD	Software Version Description (this document)
SW	Software
UI	User Interface
V&V	Verification and Validation
VCRM	Verification and Compliance Requirements Matrix
WBS	Work Breakdown Structure

APPENDIX A – CSCI Directory Contents

The following directory tree represents the complete contents of the 7200_App_Package_v20260322 delivery folder:

```
7200_App_Package_v20260322/
├── 7200_System_v2.html              (870 KB)  Main application
├── 7200_Test_System.html            (194 KB)  Test & validation suite
├── 7200_User_Guide.docx             ( 18 KB)  User manual (Hebrew)
├── server.py                        ( 19 KB)  Optional Python server
├── requirements.txt                 (  1 KB)  Python dependencies
├── start_server.bat                 (  1 KB)  Windows server launcher
├── start_server.sh                  (  1 KB)  Linux/macOS server launcher
├── data/                                       Reference data & knowledge base
└── TEMPLATES/
    ├── SW_DOCS/
    │   └── טרם הוכנס לאתר/  SDD, SDP(x3), SRD, SRS, STD, STP, STR, SUM, SVD
templates
    ├── SYSTEM_DOC/              OCD, SEMP, SSDD, SYRS, MITP, ICD, SSD, SYRS
templates
    └── Milestone_PPT_Template/  SRR, SDR, PDR, CDR, TRR PowerPoint templates
```